Table 1: Projected number of additional skilled workers needed for the deployment of wind and solar energy and investment in skills (in million Euro) by EU Member State and EU27 total in 2030. Average share of renewable energy sources in electricity generation over 2020-2022 (%)

Member State	Capacity additions		Additional jobs (000 jobs)		Re-training costs (in million EUR)		Share of renewables (*avg.2020-22*)
	GW / year	*GW / capita*	*min*	*max*	*min*	*max*	(%)
Austria	2.0	0.22	3.5	4.4	45.5	56.8	75.6
Belgium	2.4	0.21	4.2	6.1	47.9	59.9	26.7
Bulgaria	0.5	0.08	0.7	1.7	3.7	4.7	21.7
Croatia	0.4	0.09	0.2	1.2	1.1	1.3	54.3
Cyprus	0.3	0.29	0.4	0.6	1.4	1.8	14.6
Czechia	2.2	0.20	2.3	6.6	9.8	12.2	14.9
Germany	16.9	0.20	32.7	36.6	379.0	473.7	45.2
Denmark	0.1	0.01	0.2	0.8	9.7	12.1	71.8
Spain	3.5	0.07	7.2	12.9	44.2	55.3	46.6
Estonia	0.2	0.14	0.3	0.6	1.4	1.7	28.9
Finland	0.9	0.16	0.2	3.6	2.3	2.9	42.4
France	4.4	0.06	4.5	13.3	142.8	178.5	25.6
Greece	2.0	0.19	3.0	5.1	19.3	24.2	38.1
Hungary	0.4	0.04	0.6	1.4	1.7	2.2	13.6
Ireland	1.3	0.25	2.6	2.8	25.4	31.7	37.4
Italy	15.1	0.26	25.8	33.8	199.2	249.0	37.1
Lithuania	0.3	0.12	0.1	1.5	0.4	0.5	22.6
Luxembourg	0.3	0.52	0.3	1.1	3.6	4.5	14.7
Latvia	0.3	0.18	0.5	1.0	1.7	2.1	52.7
Malta	0.1	0.10	0.1	0.1	0.4	0.5	9.8
Netherlands	3.3	0.19	3.4	11.4	121.3	151.7	33.2
Poland	3.2	0.08	5.1	9.0	16.6	20.7	18.1
Portugal	1.7	0.16	2.8	3.5	11.9	14.8	59.1
Romania	2.0	0.11	2.0	6.0	4.9	6.2	43.3
Slovakia	0.5	0.09	1.0	1.3	3.9	4.9	22.8
Slovenia	0.3	0.16	0.2	1.0	1.4	1.7	35.7
Sweden	0.9	0.08	0.8	2.4	8.8	11.0	77.9
EU27	65.4	0.15	130.2	144.7	1,109.4	1,386.7	38.8

Source: Own calculations; share of renewables in electricity from Eurostat (2024b).
Notes: Number of additional jobs obtained from JRC-GEM-E3 calculations using the input-output (IO) tables projected for the year 2025 in the Global Energy and Climate Outlook 2021 (Wojtowicz et al., 2021; Keramidas et al., 2021). Training costs estimates per person per year from European Commission (2020a) and Vandeplas et al. (2022). Capacity additions, additional jobs and re-training costs for the EU27 are the sum of the individual Member States, while the share of renewables in electricity reflect the EU average over 2020-22 from Eurostat (2024b).

In general, the literature estimates employment factors for the construction and installation ranging from 2.0 to 6.2 (full-time equivalent) jobs/MW of wind power installed, and from 3.5 to 13.0 jobs/MW of solar power installed. ([66]) These estimates are typically derived from bottom-up technology models and provide a useful reference to cross-check the employment factors obtained in our Input-Output (IO) estimates. In our analysis, we find 2.4 jobs/MW installed of wind power and 2.1 jobs/MW installed of solar power when looking at the EU27 average. While the employment factors for wind are within the range observed in the literature, we note that our results for solar are conservative compared to the existing literature.

The deployment of wind and solar energy affects human capital, as more trained workers are needed to support the capacity expansion of these technologies. Following our calculations, we find that the additional installed capacities of wind and solar to deliver on

([66]) Cameron and Van der Zwaan (2015), Fragkos and Paroussos (2018), Ortega et al. (2020), Pai et al. (2021), European Commission (2022a and 2022b).